

Topic P8: Global challenges

This topic seeks to integrate learners' knowledge and understanding of physical systems and processes, with the aim of applying it to global challenges. Applications of physics can be used to help humans improve their own lives and strive to create a sustainable world for future generations, and these challenges are considered in this topic. It therefore provides opportunities to draw together the concepts covered in earlier topics, allowing synoptic treatment of the subject of physics.

P8.1 Physics on the move

Summary

Learners will use their knowledge of forces and motion to develop their ideas about how objects are affected by external factors. They will develop a better understanding of these external factors to be able to understand how the design of objects such as cars may be modified to operate more safely.

Underlying knowledge and understanding

Learners should be familiar with how forces affect motion of objects. They will also need to have a good understanding of momentum from P2.2. Learners may already have some knowledge of how vehicles are adapted to increase safety.

Common misconceptions

Learners tend to confuse the factors that affect thinking distance and braking distance, thinking that alcohol, drugs and tiredness will affect braking distance rather than thinking distance. It needs to be made clear the distinction between these two terms and that the combination of these gives us the stopping distance.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
P8.1a	recall typical speeds encountered in everyday experience for wind and sound, and for walking, running, cycling and other transportation systems	M1d		
P8.1b	estimate the magnitudes of everyday accelerations	M1d		
P8.1c	make calculations using ratios and proportional reasoning to convert units and to compute rates	conversion from non-SI to SI units	M1c, M3c	

P8.1d	explain methods of measuring human reaction times and recall typical results		M1a, M2a, M2b	WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3b, WS1.3c, WS1.3e, WS1.3g, WS1.3h, WS2a, WS2b, WS2c, WS2d	Investigation of reaction time using ruler drop experiments. (PAG P3)
P8.1e	explain the factors which affect the distance required for road transport vehicles to come to rest in emergencies and the implications for safety	factors that affect thinking and braking distance and overall stopping distance			
P8.1f ☒	estimate how the distances required for road vehicles to stop in an emergency, varies over a range of typical speeds		M1c, M1d, M2c, M2h, M3b, M3c	WS1.1e, WS1.1h	Research stopping distances using the Highway Code.
P8.1g	explain the dangers caused by large decelerations			WS1.1e, WS1.1f, WS1.1h, WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS2a, WS2b	Research and building of casing on trolleys for eggs to investigate crumple zones and safety features in cars.
P8.1h ☒	estimate the forces involved in typical situations on a public road				
P8.1i ☒	estimate, for everyday road transport, the speed, accelerations and forces involved in large accelerations		M1d, M2b, M2h, M3c		